README_TableTask_NSTE_Analysis

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This document describes the Network Symbolic Transfer Entropy (NSTE) stage of the TableTask cardiac synchrony pipeline. It explains the analysis workflow, required inputs, the purpose of each script, the adaptations made from the original laboratory implementation provided by Jackie, the selected analysis parameters, and the expected outputs. The goal is to provide sufficient documentation for Veronica and future research assistants to understand, reproduce, and maintain this stage of the pipeline.

1. Pipeline Position

Scripts 01–04 prepare the ECG recordings, derive heart-rate time series, segment them into trial-level paired signals, and compute synchrony metrics. The NSTE stage is performed after Step 4 and operates only on the paired 4 Hz heart-rate files generated by the preprocessing pipeline.

2. Scripts

05_script_tabletask_nste_v4_final_EG.py
Primary analysis script.

Main workflow:

- Automatically locate all paired trial heart-rate files.
- Read paired signals for each trial.
- Verify sampling consistency.
- Convert continuous heart-rate signals into symbolic sequences.
- Estimate directional Symbolic Transfer Entropy (STE).
- Estimate Network Symbolic Transfer Entropy (NSTE).
- Repeat calculations across all candidate delay (τ) values.
- Repeat calculations across multiple overlapping analysis windows.
- Generate surrogate distributions.
- Merge metadata from the earlier QC pipeline.
- Export standardized tables for downstream statistical analysis.

05a_script_tabletask_nste_plots_v4_final_EG.py
Reads the exported NSTE tables and generates publication-quality summary figures for each dyad. It does not recompute NSTE. It produces NSTE asymmetry plots, STE asymmetry plots, and a manifest listing every generated figure.

Supporting NSTE Functions

The main NSTE analysis script (05_script_tabletask_nste_v4_final_EG.py) relies on several supporting Python modules that implement the underlying symbolic transfer entropy calculations. These files are adapted from the original laboratory NSTE implementation and are called automatically during the analysis. They should remain in the same directory as the main script and generally do not need to be run individually.

- **calculate_STE.py**: Computes Symbolic Transfer Entropy (STE) between two symbolic time series and provides the directional information transfer estimates used by the NSTE pipeline.
- **delayRecons.py**: Performs delay-coordinate reconstruction of the symbolic time series by constructing embedding vectors from past observations before transfer entropy estimation.
- **F_EstimateProb.py**: Estimates the joint and conditional probability distributions required for entropy and transfer entropy calculations.
- **F_Int2Symb.py**: Converts continuous-valued heart-rate signals into symbolic sequences that form the basis of the symbolic transfer entropy analysis.
- **F_Integer2prob.py**: Converts encoded symbolic states into probability distributions used during entropy estimation.
- **f_nste.py**: Implements the core Network Symbolic Transfer Entropy (NSTE) algorithm by combining symbolic encoding, state-space reconstruction, probability estimation, and transfer entropy calculations into the final directional NSTE estimates.
- **f_predictiontime.py**: Determines the prediction delay used during transfer entropy estimation by evaluating candidate prediction horizons, following the methodology implemented in the original laboratory pipeline.

These supporting modules implement the mathematical components of the NSTE algorithm, while the main analysis script manages data loading, trial processing, metadata integration, surrogate testing, output generation, and reproducible analysis of the TableTask dataset.

3. Adaptation from Jackie's Original Laboratory Scripts

The objective of the adaptation was to preserve the original NSTE methodology while making the implementation compatible with the TableTask project. The statistical estimation procedure was not intentionally redesigned. Most modifications concern automation, reproducibility, and integration with the existing preprocessing pipeline.

Component	**Original script**	**TableTask adaptation**
NSTE estimator	Preserved	Core computational procedure retained.
STE calculation	Preserved	Directional STE retained.
Symbolization	Preserved	Original symbolic representation retained.
Surrogate methodology	Preserved	Same surrogate-testing logic retained.

Parameter defaults	Preserved	Default analysis settings retained unless required for integration.
Input discovery	Manual	Automatically detects paired HR files from the Step 3 output directory.
Metadata	Minimal	Automatically merges analysis-unit and RR reliability metadata.
Directory structure	Project-specific	Uses standardized TableTask folder hierarchy.
Output naming	Project-specific	Standardized filenames and folder organization.
Logging	Limited	Run summaries, failure logs and manifests added for reproducibility.
Plot generation	Embedded/combined	Separated into a dedicated plotting script.

4. Parameters

The following parameters were retained from the original laboratory implementation unless noted otherwise.

Parameter	**Value**	**Origin**	**Comment**
Sampling frequency	4 Hz	TableTask	Determined by Step 3 interpolation.
Embedding dimension	3	Jackie	Adopted from original implementation. Publication-specific optimization should be confirmed by Veronica.
τ search	1–13 samples	Jackie	Adopted directly.
Window lengths	10,15,30,45,60 s	Jackie	Adopted directly.

Window overlap	80%	Jackie	Adopted directly.
Internal shuffles	20	Jackie	Adopted directly.
Surrogates	400	Jackie	Adopted directly for final analysis.
Random seed	448	Added	Introduced only for reproducibility.

5. Inputs

Required inputs:

- processed_ecg_hr/trial_segments_4hz/paired_trial_hr/
- processed_ecg_hr/trial_segments_4hz/qc/03_trial_segmentation_summary.csv
- processed_ecg_hr/rr_reliability_qc/qc/3a_trial_dyad_rr_reliability.csv
- qc_outputs/veronica_analysis_units.csv

6. Outputs

Script 05 creates a trial_nste_final_EG directory containing:

- Window-level NSTE tables
- Trial summary tables
- Run summary
- Failure log
- Plot manifest

Script 05a creates:

- NSTE asymmetry figures
- STE asymmetry figures

7. Notes

The adaptation was designed to preserve the statistical methodology used in the original laboratory implementation while making the workflow reproducible and compatible with the TableTask preprocessing pipeline. Any future methodological changes, including embedding dimension, surrogate count, window configuration, or inferential procedures, should be discussed with Veronica before they are adopted for a publication-quality analysis.